

1. (a) Find z , which satisfies $a \oplus z = z \oplus a = a, \forall a \in \mathbb{Z}$
 i.e. $a + z - 1 = z + a - 1 = a \Rightarrow z = 1$

(b) Find b , which satisfies $a \oplus b = b \oplus a = z, \forall a \in \mathbb{Z}$
 i.e. $a + b - 1 = 1 \Rightarrow b = 2 - a$

(c) $\forall a, b \in \mathbb{Z}$, consider that $a \oplus b = a + b - 1 = b + a - 1 = b \oplus a$
 Thus, $\oplus$ commutes.

(d) Find u , which satisfies $a \odot u = u \odot a = a, \forall a \in \mathbb{Z}$ and $u \neq z$
 i.e. $a + u - au = u + a - ua = a \Rightarrow u(1-a) = 0$ for any $a \in \mathbb{Z} \Rightarrow u = 0$

2. (a) Consider that $T_0 = 2, T_1 = 3, T_2 = 6, T_n = (n+4)T_{n-1} - 4^n T_{n-2} + (4n-8)T_{n-3}$

$$\text{Then, } T_3 = (3+4) \times 6 - 4 \times 3 \times 3 + (4 \times 3 - 8) \times 2 = 42 - 36 + 8 = 14$$

$$T_4 = (4+4) \times 14 - 4 \times 4 \times 6 + (4 \times 4 - 8) \times 3 = 112 - 96 + 24 = 40$$

Find that $T_0 = 2 = 1+1 = 2^0 + 0!$

$$T_1 = 3 = 2+1 = 2^1 + 1!$$

$$T_2 = 6 = 4+2 = 2^2 + 2!$$

$$T_3 = 14 = 8+6 = 2^3 + 3!$$

$$T_4 = 40 = 16+24 = 2^4 + 4!$$

Thus, we suppose that $T_n = 2^n + n!$

(b) Using induction:

$$\text{When } n=0: T_0 = 2^0 + 0! = 2 \quad \checkmark$$

$$\text{When } n=1: T_1 = 2^1 + 1! = 3 \quad \checkmark$$

$$\text{When } n=2: T_2 = 2^2 + 2! = 6 \quad \checkmark$$

When $n \geq 2$: Suppose that $T_k = 2^k + k!$ for any $k \leq n$

$$T_{n+1} = (n+5)T_n - 4(n+1)T_{n-1} + (4n-4)T_{n-2}$$

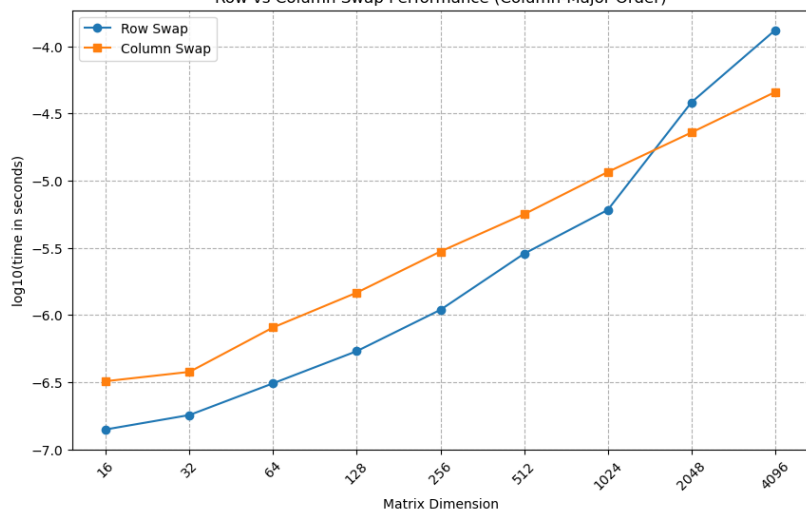
$$= (n+5)(2^n + n!) - 4(n+1)(2^{n-1} + (n-1)!) + (4n-4)(2^{n-2} + (n-2)!)$$

$$= n \cdot 2^n + n \cdot n! + 5 \cdot 2^n + 5 \cdot n! - 2n \cdot 2^n - 4n! - 2 \cdot 2^n - 4(n-1)! + n \cdot 2^n + 4(n-1)! - 2^n$$

$$= 2 \cdot 2^n + n \cdot n! + n! = 2^{n+1} + (n+1)!$$

Thus, the formula works

Row vs Column Swap Performance (Column-Major Order)



4. Consider for what C_1, C_2, C_3 .

$$C_1 \cdot 1 + C_2 \cdot (1-x) + C_3 \cdot (1-x)^2 = 0$$

$$\text{Then } C_1 + C_2 - C_2x + C_3x^2 - 2C_3x + C_3 = 0$$

$$\Rightarrow C_3x^2 - (C_2 + 2C_3)x + C_1 + C_2 + C_3 = 0 \quad \text{for any } x$$

$$\Rightarrow \begin{cases} C_3 = 0 \\ C_2 + 2C_3 = 0 \\ C_1 + C_2 + C_3 = 0 \end{cases} \Rightarrow \begin{cases} C_1 = 0 \\ C_2 = 0 \\ C_3 = 0 \end{cases}$$

Therefore, $1, 1-x, (1-x)^2$ are linearly independent.

What's more, for any $f(x)$, which is polynomials to degree 2

$$\text{Consider } f(x) = ax^2 + bx + c$$

$$\text{Let } \begin{cases} a = C_3 \\ b = -(C_2 + 2C_3) \\ c = C_1 + C_2 + C_3 \end{cases}, \text{ then } f(x) = ax^2 + bx + c = C_3(1-x)^2 + C_2(1-x) + C_1$$

Thus, $\{1, 1-x, (1-x)^2\}$ span the space

Or consider that

$$(1 \ 1-x \ (1-x)^2) = (1 \ x \ x^2) \begin{pmatrix} 1 & 1 & 1 \\ 0 & -1 & -2 \\ 0 & 0 & 1 \end{pmatrix}$$

Since, $1, x, x^2$ are linearly independent and let $A = \begin{pmatrix} 1 & 1 & 1 \\ 0 & -1 & -2 \\ 0 & 0 & 1 \end{pmatrix}$

$$\det(A) = -1 \neq 0, \text{ rank } A = 3$$

Thus, $(1, 1-x, (1-x)^2)$ are linearly independent and span the space

(b) Using the Gram-Schmidt:

$$\text{Let } x_1'(t) = t^2$$

$$\text{Then } x_2'(t) = x_2 - \frac{(x_2, x_1')}{(x_1', x_1')} \cdot x_1'$$

$$= t - \frac{\int_{-1}^1 t \cdot t^2 dt}{\int_{-1}^1 t^2 \cdot t^2 dt} \cdot t^2 = t - \frac{\frac{t^4}{4} \Big|_{-1}^1}{\frac{t^5}{5} \Big|_{-1}^1} \cdot t^2 = t$$

$$\begin{aligned} \chi_3'(t) &= \chi_3 - \frac{(\chi_3, \chi_1')}{(\chi_1', \chi_1')} \cdot \chi_1' - \frac{(\chi_3, \chi_2')}{(\chi_2', \chi_2')} \cdot \chi_2' \\ &= 1 - \frac{\int_{-1}^1 1 \cdot t^2 dt}{\int_{-1}^1 t^2 \cdot t^2 dt} \cdot t^2 - \frac{\int_{-1}^1 1 \cdot t dt}{\int_{-1}^1 t \cdot t dt} \cdot t \\ &= 1 - \frac{\frac{t^3}{3} \Big|_{-1}^1}{\frac{t^5}{5} \Big|_{-1}^1} \cdot t^2 - \frac{\frac{t^2}{2} \Big|_{-1}^1}{\frac{t^3}{3} \Big|_{-1}^1} \cdot t = 1 - \frac{2}{3} t^2 \end{aligned}$$

Thus, the orthonormal basis can be $\{t^2, t, 1 - \frac{2}{3}t^2\}$

5.

